

Notes: Staiger & Stock (Econometrica, 1997)

Instrumental Variables Regression with Weak Instruments

Contents

1	Big picture	3
2	Data, notation, and measurement	3
2.1	Structural equation and first stage	3
2.2	Weak-instrument local-to-zero assumption	3
2.3	Concentration parameter and first-stage strength	4
2.4	Correlation between structural and first-stage errors	4
2.5	Empirical application data	4
3	Models and empirical specifications	4
3.1	Why conventional asymptotics fail	4
3.2	The k -class estimator	5
3.3	Limiting behavior of TSLS and LIML	5
3.4	Bias of TSLS relative to OLS	5
3.5	Overidentification tests	5
3.6	Exogeneity tests	6
3.7	Anderson–Rubin confidence regions	6
3.8	Bonferroni confidence regions	6
4	Identification and theoretical design	6
4.1	What is being identified?	6
4.2	Why large samples do not solve the problem	6
4.3	Why the local-to-zero sequence is credible	7
4.4	What the Monte Carlo section validates	7
4.5	What the paper cannot fully solve	7
5	Tables and figures	8
5.1	Figure 1: Asymptotic and finite-sample pdfs of the LIML estimator and t statistic	8
5.2	Table I: Maximum absolute difference between finite-sample CDF and asymptotic CDF	8
5.3	Figure 2: Asymptotic bias of TSLS and asymptotic median bias of LIML	9
5.4	Figure 3: Asymptotic coverage rates of conventional 95% TSLS and LIML confidence intervals	10
5.5	Table II: Estimated effects of years of education on log weekly earnings in the 1980 Census	11
6	Short summary	13
7	Conclusion	13

7.1	Core conclusion	13
7.2	Economic intuition	13
7.3	Practical implications for empirical researchers	13
7.4	How to read the Angrist–Krueger application	14
7.5	Broader takeaway	14
7.6	Limitations and cautions	14

1 Big picture

- **Question.** What happens to IV estimators, tests, and confidence intervals when the instruments are weakly correlated with the endogenous variables?
- **Core problem.** Standard IV asymptotics treat first-stage coefficients as fixed. Then the first-stage F statistic diverges as sample size grows. This is a bad approximation when first-stage F is small or moderate.
- **Key asymptotic device.** The paper models first-stage coefficients as local to zero, so first-stage strength stays finite as $T \rightarrow \infty$.
- **Main theoretical result.** Under weak-instrument asymptotics, TSLS, LIML, Wald statistics, t statistics, overidentification tests, and exogeneity tests have nonstandard limiting distributions.
- **Main practical result.** TSLS can be strongly biased toward OLS and conventional TSLS confidence intervals can have severe undercoverage, even in very large samples.
- **LIML result.** LIML is much less biased than TSLS in many weak-instrument designs, especially when the concentration parameter per instrument is not too small.
- **Inference result.** Anderson–Rubin and Bonferroni confidence regions are more reliable under weak instruments than conventional Wald intervals.
- **Empirical application.** Reinterpreting Angrist and Krueger’s quarter-of-birth instruments, the paper argues that reliable LIML estimates of returns to education are around 8–10%, with wider confidence intervals than conventional TSLS suggests.
- **Economic/econometric takeaway.** Large sample size does not by itself solve weak identification. What matters is first-stage strength, summarized by the concentration parameter and related bias measures.

2 Data, notation, and measurement

2.1 Structural equation and first stage

The paper studies a single-equation IV model with endogenous regressors:

$$y = Y\beta + X\gamma + u, \tag{1}$$

$$Y = ZH + X\Phi + V. \tag{2}$$

- y is a $T \times 1$ dependent variable.
- Y is a $T \times n$ matrix of endogenous regressors.
- X is a $T \times K_1$ matrix of included exogenous regressors.
- Z is a $T \times K_2$ matrix of excluded instruments.
- H is the matrix of first-stage coefficients on the excluded instruments.

Interpretation: The problem is not that instruments are invalid. The problem is that valid instruments can be too weakly correlated with Y for standard asymptotic approximations to be useful.

2.2 Weak-instrument local-to-zero assumption

The central asymptotic assumption is

$$H = H_T = \frac{C}{\sqrt{T}}, \tag{3}$$

where C is fixed as $T \rightarrow \infty$.

Interpretation: If H were fixed, the first-stage F statistic would diverge. By shrinking H at rate $T^{-1/2}$, the paper keeps the first-stage signal finite, matching the empirical case in which F is small even in large samples.

2.3 Concentration parameter and first-stage strength

A central object is the concentration parameter matrix, often summarized by

$$\frac{A'A}{K_2}. \quad (4)$$

For $n = 1$, $A'A/K_2$ is closely related to the expected first-stage F statistic.

Interpretation: A small value of $A'A/K_2$ means weak identification. A large sample with small $A'A/K_2$ behaves more like a weakly identified finite-sample problem than like a conventional large-sample IV problem.

2.4 Correlation between structural and first-stage errors

Let ρ denote the correlation between the structural error and the first-stage error after normalization. When $|\rho|$ is large, the OLS bias is large and weak-instrument TSLS is especially dangerous.

Interpretation: Weak instruments matter most when endogeneity is substantial. If OLS and IV estimands are close, weak instruments have less scope to cause dramatic estimator bias.

2.5 Empirical application data

The application reexamines Angrist and Krueger’s returns-to-education design:

- Data are from the 1980 U.S. Census.
- Instruments are quarter-of-birth indicators and interactions with year of birth or state of birth.
- The sample is split into men born in 1930–39 and men born in 1940–49.
- Some specifications have only 3 instruments; others have 30, 28, or 178 instruments.

Interpretation: The application is ideal for the paper’s method because it combines huge samples with small first-stage F statistics in some specifications.

3 Models and empirical specifications

3.1 Why conventional asymptotics fail

Under conventional asymptotics, H is fixed, so the first-stage F statistic tends to infinity. The model therefore behaves as if instruments are strong in the limit.

Under weak-instrument asymptotics,

$$H_T = \frac{C}{\sqrt{T}}, \quad (5)$$

so the first-stage signal remains finite. This makes the IV estimator’s limit nonstandard.

Interpretation: The weak-instrument asymptotic sequence is designed to approximate the finite-sample situation that matters in applications: the first stage does not get stronger just because the sample is large.

3.2 The k -class estimator

The paper uses the k -class estimator to cover TSLS and LIML:

$$\hat{\delta}(k) = (W'(I - kM_{\bar{Z}})W)^{-1} W'(I - kM_{\bar{Z}})y, \quad (6)$$

where $W = [Y, X]$ and $\bar{Z} = [Z, X]$.

- $k = 1$ corresponds to TSLS.
- $k = k_{\text{LIML}}$ corresponds to LIML.

Interpretation: TSLS and LIML are not asymptotically equivalent under weak instruments. That is a key difference from conventional strong-instrument asymptotics.

3.3 Limiting behavior of TSLS and LIML

The paper derives joint weak-instrument limits for

- $\hat{\beta}_{\text{TSLS}}$,
- $\hat{\beta}_{\text{LIML}}$,
- Wald statistics,
- t statistics,
- overidentification tests,
- exogeneity tests.

Interpretation: The distributions depend on nuisance parameters such as $A'A/K_2$ and ρ . Because $A'A/K_2$ is not consistently estimable under weak-instrument asymptotics, usual plug-in normal inference is invalid.

3.4 Bias of TSLS relative to OLS

For $n = 1$, the paper gives a simple bias diagnostic:

$$B_{\max} = \left(\frac{G_T}{K_2} \right)^{-1}, \quad (7)$$

where G_T/K_2 is the first-stage F statistic.

Interpretation: For one endogenous regressor, the inverse first-stage F statistic approximates the relative bias of TSLS compared with OLS. This is the intuition behind the paper's practical focus on reporting first-stage F statistics.

3.5 Overidentification tests

The paper studies Basmann-type overidentification tests based on TSLS and LIML. Under weak instruments, asymptotically equivalent strong-IV tests are no longer equivalent.

Interpretation: The TSLS-based Basmann test can severely overreject when instruments are weak and $|\rho|$ is large. The LIML-based version has better size but can be conservative.

3.6 Exogeneity tests

The paper studies Durbin–Wu–Hausman-type tests of exogeneity. It finds that some forms are conservative, while the Durbin form has correct asymptotic size and useful power.

Interpretation: A weak first stage affects not only estimation of β but also tests comparing OLS and IV estimands.

3.7 Anderson–Rubin confidence regions

The Anderson–Rubin test inverts a statistic that remains valid under weak instruments. It tests values of β by asking whether the instruments explain the residualized structural equation.

Interpretation: AR intervals are robust to weak instruments, but can be wide or empty depending on overidentification and model fit.

3.8 Bonferroni confidence regions

The paper constructs Bonferroni regions by combining:

1. a confidence set for the concentration parameter $A'A/K_2$;
2. a conditional confidence region for β given that concentration parameter.

Interpretation: Bonferroni intervals are conservative by construction but can be more informative than AR intervals when the number of instruments is large.

4 Identification and theoretical design

4.1 What is being identified?

The structural parameter β is identified when instruments are valid and sufficiently relevant. Weak instruments create weak identification, meaning that the data contain little information about β relative to the noise in the first stage.

Interpretation: The paper does not say IV is invalid. It says that standard IV inference can be unreliable when instrument relevance is weak.

4.2 Why large samples do not solve the problem

The Angrist–Krueger application has hundreds of thousands of observations, yet some first-stage F statistics are below 5. Weak-instrument asymptotics explain why conventional inference can still fail.

Interpretation: Identification strength is not equal to sample size. The information relevant for IV comes from the strength of the excluded instruments after controls.

4.3 Why the local-to-zero sequence is credible

Modeling $H = C/\sqrt{T}$ keeps the expected first-stage statistic finite. This directly approximates empirical designs with low first-stage F .

Interpretation: The asymptotics are not intended to describe the true data-generating process exactly. They are a tool for approximating weakly identified finite-sample behavior.

4.4 What the Monte Carlo section validates

The simulations show that the weak-IV asymptotic distributions approximate finite-sample distributions well when there are roughly 10–20 observations per instrument.

Interpretation: This gives applied researchers a practical reason to use the weak-IV approximations rather than relying purely on exact finite-sample formulas.

4.5 What the paper cannot fully solve

- The concentration parameter is not consistently estimable under the weak-instrument asymptotics.
- No single weak-IV confidence procedure dominates uniformly.
- The paper focuses on a fixed number of instruments and endogenous variables.
- The recommendations depend on the pattern of first-stage strength, number of instruments, and endogeneity correlation.

Interpretation: The paper provides practical diagnostics and robust procedures, but not a single universal replacement for all IV inference.

5 Tables and figures

5.1 Figure 1: Asymptotic and finite-sample pdfs of the LIML estimator and t statistic

Read:

- The figure compares asymptotic pdfs with finite-sample pdfs in two designs.
- Solid lines are weak-instrument asymptotic approximations; dashed lines are finite-sample distributions.
- Panels (a) and (b) focus on LIML estimator distributions.
- Panels (c) and (d) focus on the LIML t statistic.
- The finite-sample and asymptotic densities line up closely in the displayed cases.
- The point is not that the distributions are normal; it is that the nonstandard weak-IV approximation captures the finite-sample shape.

Economic message: Weak-instrument asymptotics are useful because they match the shape of finite-sample IV distributions in cases where conventional normal approximations fail.

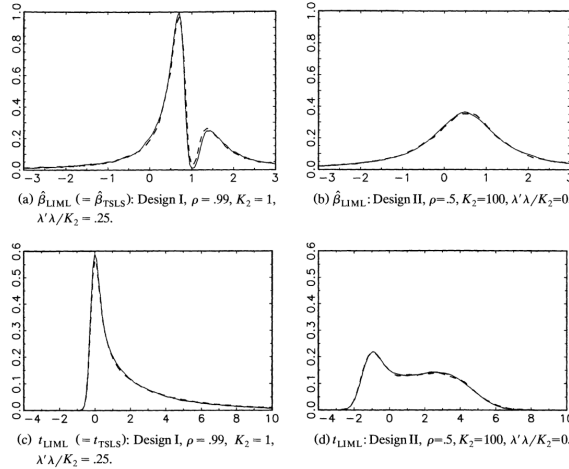


FIGURE 1—Asymptotic (solid line) and finite-sample (dashed line) pdf's of the LIML estimator and t statistic. True $\beta_0 = 0$; $\text{plim}(\hat{\beta}_{OLS}) = \rho$.

Figure 1: Asymptotic and finite-sample pdfs of the LIML estimator and t statistic.

5.2 Table I: Maximum absolute difference between finite-sample CDF and asymptotic CDF

Read:

- The table reports Kolmogorov–Smirnov differences between finite-sample and weak-IV asymptotic CDFs.
- The designs vary the number of instruments K_2 , the error correlation ρ , the concentration parameter $A'A/K_2$, and observations per instrument T/K_2 .
- Differences generally shrink as T/K_2 rises from 5 to 10 to 20.

- For estimator distributions, the approximation is already quite good with 10–20 observations per instrument.
- The t statistic approximations are somewhat less accurate than estimator approximations, but still improve substantially with T/K_2 .
- The table supports using weak-IV asymptotics beyond the fixed-instrument Gaussian environment.

Economic message: The weak-IV asymptotic approximation is not merely a theoretical curiosity; it works in the simulation designs that mimic common time-series and cross-sectional IV applications.

574

D. STAIGER AND J. H. STOCK

TABLE I
MAXIMUM ABSOLUTE DIFFERENCE BETWEEN FINITE SAMPLE CDF AND ASYMPTOTIC CDF^a

Parameters		$T/K_2 = 5$					$T/K_2 = 10$					$T/K_2 = 20$				
K_2	ρ	$\lambda A/K_2$	$\hat{\beta}_{\text{TSL}}$	t_{TSL}	$\hat{\beta}_{\text{LIML}}$	t_{LIML}	$\hat{\beta}_{\text{TSL}}$	t_{TSL}	$\hat{\beta}_{\text{LIML}}$	t_{LIML}	$\hat{\beta}_{\text{TSL}}$	t_{TSL}	$\hat{\beta}_{\text{LIML}}$	t_{LIML}	$\hat{\beta}_{\text{TSL}}$	t_{TSL}
A. Design I																
1	.99	0	0.007	0.075	0.007	0.075	0.008	0.029	0.008	0.029	0.004	0.012	0.004	0.012		
1	.99	.25	0.089	0.093	0.089	0.093	0.047	0.046	0.047	0.046	0.025	0.022	0.025	0.022		
1	.99	1	0.111	0.087	0.111	0.087	0.050	0.038	0.050	0.038	0.026	0.026	0.026	0.026		
1	.99	10	0.081	0.090	0.081	0.090	0.037	0.059	0.037	0.059	0.021	0.045	0.021	0.045		
4	.99	0	0.007	0.052	0.008	0.025	0.006	0.024	0.005	0.013	0.005	0.008	0.005	0.006		
4	.99	.25	0.041	0.135	0.032	0.027	0.023	0.085	0.016	0.016	0.012	0.048	0.015	0.015		
4	.99	1	0.063	0.058	0.030	0.051	0.031	0.041	0.012	0.034	0.021	0.024	0.009	0.024		
4	.99	10	0.033	0.015	0.015	0.012	0.022	0.010	0.010	0.008	0.010	0.006	0.011	0.009		
B. Design II																
4	.5	0	0.041	0.080	0.028	0.038	0.027	0.044	0.017	0.021	0.014	0.023	0.007	0.014		
4	.5	.25	0.051	0.061	0.038	0.046	0.025	0.039	0.020	0.024	0.015	0.018	0.013	0.018		
4	.5	1	0.059	0.039	0.037	0.038	0.034	0.024	0.024	0.023	0.015	0.014	0.013	0.016		
4	.5	10	0.067	0.051	0.058	0.060	0.041	0.031	0.041	0.043	0.025	0.019	0.025	0.026		
100	.5	0	0.050	0.048	0.011	0.019	0.029	0.026	0.012	0.011	0.020	0.015	0.004	0.009		
100	.5	.25	0.052	0.045	0.021	0.020	0.031	0.028	0.012	0.013	0.016	0.016	0.013	0.013		
100	.5	1	0.037	0.034	0.034	0.034	0.027	0.028	0.025	0.026	0.021	0.020	0.022	0.022		
100	.5	10	0.067	0.069	0.077	0.077	0.048	0.047	0.050	0.050	0.037	0.038	0.042	0.042		

^a Entries are the Kolmogorov-Smirnov statistics testing the equality of the two distributions; specifically, $\sup_x |F_{\text{asy}}(x) - F_{\text{exact}}(x)|$, where F_{asy} and F_{exact} are the Monte Carlo estimates of the asymptotic and exact finite sample distributions. Quantiles of this statistic, under the hypothesis that the two population distributions are identical, are: 50%, .0064; 95%, .0105. Asymptotic distributions were computed using 100,000 replications of the representation in Theorems 1 and 2. Finite sample distributions were computed using 20,000 replications.

statistic for TSLS and LIML are somewhat less good, but for $T/K_2 = 10$ they are typically within .03 in both designs.

Finite-sample coverage rates of 95% Bonferroni (both TSLS and LIML, with $\alpha_1 = \alpha_2 = .025$) and AR confidence intervals were checked for the models in Table I. For Bonferroni intervals, the lowest coverage rate for $T/K_2 = 10$ is 93.4%; for $T/K_2 = 20$ coverage is at least 95% and typically is between 96% and 99%. The AR interval coverage rates are between 93% and 95% for $T/K_2 = 20$.⁸

These results suggest that the weak instrument asymptotics provide good approximations to the finite sample distributions of the estimators and t statistics when exogenous regressors are stochastic and errors are nonnormal, for a wide range of parameter values including cases previously studied by Nelson and Startz (1990a,b) and Bound, Jaeger, and Baker (1995). The approximations are

⁸ Details of these and other unreported results are available from the authors upon request.

Table I: Maximum absolute difference between finite-sample CDF and asymptotic CDF.

5.3 Figure 2: Asymptotic bias of TSLS and asymptotic median bias of LIML

Read:

- Figure 2 plots TSLS asymptotic bias and LIML median bias as a fraction of OLS bias.
- The horizontal axes vary the number of instruments K_2 and concentration strength $A'A/K_2$.
- TSLS relative bias depends strongly on $A'A/K_2$ and can remain large when there are many instruments.
- LIML becomes approximately median unbiased more quickly as $A'A/K_2$ increases.
- The figure shows why simply adding many weak instruments can make TSLS look more like OLS.
- The paper notes that for $A'A/K_2 \geq 2$, LIML median bias is much smaller than TSLS bias in many of the displayed designs.

Economic message: The central estimator lesson is: weak instruments pull TSLS toward OLS, while LIML is much more robust to this bias in many designs.

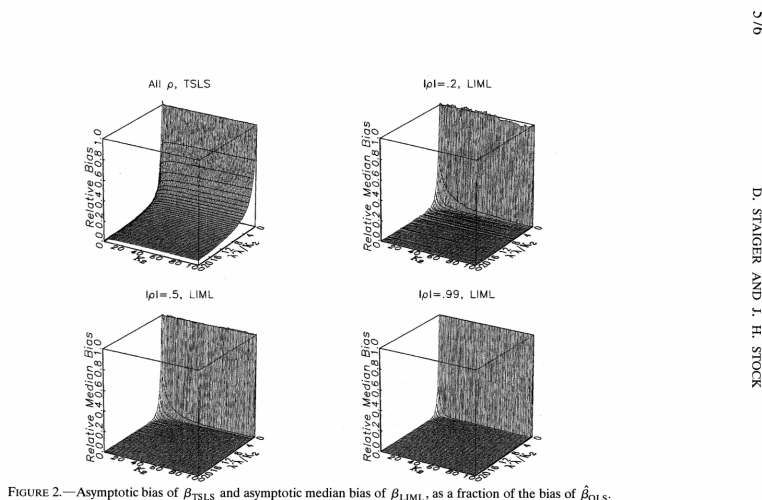


Figure 2: Asymptotic bias of TSLS and asymptotic median bias of LIML, as a fraction of OLS bias.

5.4 Figure 3: Asymptotic coverage rates of conventional 95% TSLS and LIML confidence intervals

Read:

- Figure 3 shows coverage of conventional 95% confidence intervals.
- TSLS coverage is highly sensitive to K_2 , $|\rho|$, and $A'A/K_2$.
- Coverage falls as instruments become weaker, as the number of instruments grows, and as endogeneity becomes stronger.
- TSLS intervals can undercover severely even in parameter regions that might not look obviously disastrous from sample size alone.
- LIML conventional intervals perform better than TSLS conventional intervals, but they are not uniformly reliable.
- The figure motivates nonstandard AR and Bonferroni confidence regions.

Economic message: Weak instruments do not just bias point estimates; they invalidate the standard errors and confidence intervals researchers usually report.

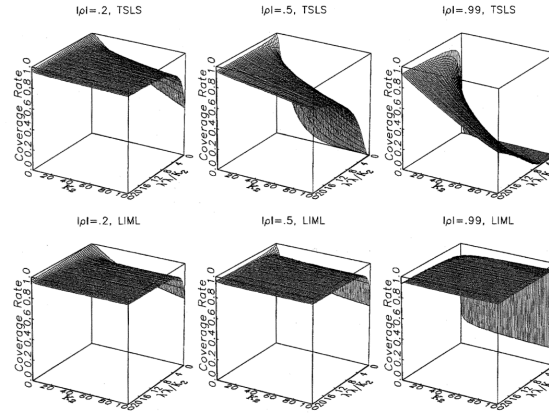


FIGURE 3.—Asymptotic coverage rates of conventional 95% TSLS and LIML confidence intervals.

Figure 3: Asymptotic coverage rates of conventional 95% TSLS and LIML confidence intervals.

5.5 Table II: Estimated effects of years of education on log weekly earnings in the 1980 Census

Read:

- Table II reexamines Angrist and Krueger's quarter-of-birth design for returns to education.
- The 1930–39 cohort has 329,509 observations; the 1940–49 cohort has 486,926 observations.
- Specification I uses 3 quarter-of-birth instruments and has a strong first stage: $F = 30.53$ for the 1930–39 cohort and $F = 26.32$ for the 1940–49 cohort.
- Specifications II–IV add many interactions, increasing the number of instruments to 30, 28, and 178.
- As the instrument list expands, some first-stage F statistics become very low, especially in specification IV where F is about 1.9.
- For the 1930–39 cohort, LIML estimates in reliable specifications are around 8–10%.
- For the 1940–49 cohort, the paper focuses on the specification with 178 instruments after considering overidentification tests and weak-IV reliability.
- AR and Bonferroni confidence intervals are often much wider than conventional TSLS intervals.

Economic message: The empirical takeaway is that weak-instrument diagnostics change the interpretation of a famous IV application: the preferred returns-to-education estimates are higher and less tightly estimated than conventional TSLS suggests.

TABLE II
ESTIMATED EFFECTS OF YEARS OF EDUCATION ON LOG WEEKLY EARNINGS IN THE 1980 CENSUS

	I	II	III	IV
<i>A. Men Born 1930–39 (n = 329, 509)</i>				
OLS	.0632	.0632	.0632	.0628
(S.E.)	(.0003)	(.0003)	(.0003)	(.0003)
TSLs	.0990	.0806	.0600	.0811
(S.E.)	(.0207)	(.0164)	(.0290)	(.0109)
LIML	.0999	.0838	.0574	.0982
(S.E.)	(.0210)	(.0179)	(.0385)	(.0153)
A-R Confidence Interval	[.052, .153]	[−.003, .179]	[−.441, .490]	[−.015, .240]
TSLs Bonferroni	[.052, .152]	[.038, .137]	[−∞, +∞]	[.048, .172]
Confidence Interval				
LIML Bonferroni	[.052, .153]	[.036, .134]	[−∞, +∞]	[.043, .158]
Confidence Interval				
F (first stage)	30.53	4.747	1.613	1.869
(p-value)	(.000)	(.000)	(.021)	(.000)
Durbin test, TSLs	3.087	1.126	0.013	2.853
(p-value)	(0.079)	(.289)	(.910)	(.091)
Basmann test, LIML	2.318	22.45	19.55	161.1
(p-value)	(.314)	(.801)	(.849)	(.800)
<i>B. Men Born 1940–49 (n = 486, 926)</i>				
OLS	.0520	.0520	.0520	.0516
(S.E.)	(.0003)	(.0003)	(.0003)	(.0003)
TSLs	−.0734	.0393	.0779	.0666
(S.E.)	(.0273)	(.0145)	(.0239)	(.0113)
LIML	−.0902	.0286	.1243	.0878
(S.E.)	(.0301)	(.0197)	(.0420)	(.0178)
A-R Confidence Interval	[∅]	[∅]	[∅]	[.033, .148]
Interval				
TSLs Bonferroni	[−.155, −.018]	[−.004, .076]	[.000, .219]	[.027, .150]
Confidence Interval				
LIML Bonferroni	[−.174, −.028]	[−.023, .079]	[−.009, .290]	[.023, .156]
Confidence Interval				
F (first stage)	26.32	6.849	2.736	1.929
(p-value)	(.000)	(.000)	(.000)	(.000)
Durbin test, TSLs	28.90	0.780	1.188	1.780
(p-value)	(.000)	(.377)	(.276)	(.182)
Basmann test, LIML	9.356	93.29	49.22	200.36
(p-value)	(.009)	(.000)	(.006)	(.110)
<i>Controls</i>				
Race, Standard Metropolitan Statistical Area, Married, Region, Year of Birth Dummies	yes	yes	yes	yes
Age, Age ²	no	no	yes	yes
State of Birth	no	no	no	yes
<i>Instruments</i>				
Quarter of birth	yes	yes	yes	yes
Quarter of birth *(year of birth)	no	yes	yes	yes
Quarter of birth *(state of birth)	no	no	no	yes
# Instruments	3	30	28	178

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Table II: Estimated effects of years of education on log weekly earnings in the 1980 Census.

6 Short summary

- The paper introduces weak-instrument asymptotics for IV regression.
- The first-stage coefficient matrix is modeled as $H_T = C/\sqrt{T}$, keeping the first-stage signal finite.
- Under this sequence, TSLS and LIML have nonstandard limiting distributions.
- TSLS can be badly biased toward OLS when instruments are weak.
- Conventional TSLS confidence intervals can have severe undercoverage.
- LIML is typically less biased than TSLS, especially when the concentration parameter per instrument is moderate.
- Standard overidentification and exogeneity tests behave differently under weak instruments than under strong instruments.
- The first-stage F statistic and related bias measures should be reported in applied work.
- AR and Bonferroni confidence regions are more reliable than conventional Wald intervals in weak-IV settings.
- The Monte Carlo evidence shows that weak-IV asymptotics approximate finite-sample distributions well with about 10–20 observations per instrument.
- In the Angrist–Krueger application, weak-IV methods point to returns to education around 8–10%, with wider intervals than conventional TSLS.
- The general lesson is that large sample size cannot rescue weak identification.

7 Conclusion

7.1 Core conclusion

Staiger and Stock show that IV inference changes fundamentally when instruments are weak. Standard asymptotic normal approximations are built for sequences in which the first-stage signal grows without bound. When the first stage remains weak, TSLS is not centered where researchers expect, t statistics are not standard normal, and Wald intervals do not have their advertised coverage.

7.2 Economic intuition

The intuition is that a weak instrument provides little independent variation in the endogenous regressor. TSLS then uses a noisy first-stage prediction, so the estimator inherits finite-sample-like behavior even as sample size grows. If the structural error and first-stage error are correlated, this noise pulls TSLS toward the biased OLS estimand. Adding many weak instruments can make the problem worse because the fitted first stage begins to absorb noise rather than true exogenous variation.

7.3 Practical implications for empirical researchers

- Always report first-stage strength: first-stage F statistics, and for multiple endogenous variables, canonical-correlation or bias measures.
- Treat a small first-stage F as an inference problem, not merely a nuisance.
- Do not rely on conventional TSLS standard errors after merely pretesting instrument relevance.
- Prefer LIML over TSLS for point estimation when instruments are weak and there is one endogenous regressor.

- Use AR or Bonferroni confidence regions when weak instruments are plausible.
- Interpret overidentification tests carefully: TSLS-based versions can overreject, while LIML-based versions can be conservative.

7.4 How to read the Angrist–Krueger application

The paper does not overturn the quarter-of-birth design. Instead, it changes the inferential standard. Some specifications have low first-stage strength despite enormous sample size. In those cases, conventional TSLS intervals are misleadingly tight. The authors’ preferred estimates are more stable when using LIML and weak-IV robust confidence methods, but the intervals are wider. This is a lesson about uncertainty: weak instruments can make apparently precise IV estimates unreliable.

7.5 Broader takeaway

The paper’s lasting contribution is to move weak instruments from a finite-sample inconvenience to a core asymptotic identification problem. It explains why applied researchers should diagnose first-stage strength and why robust confidence methods are necessary when instruments are weak. In modern empirical work, the phrase “weak instruments” is tied directly to the logic developed here: IV validity is not enough; relevance must be strong enough to support reliable inference.

7.6 Limitations and cautions

The paper provides a powerful approximation, but no procedure is uniformly best. AR intervals are robust but can be wide or empty. Bonferroni intervals can be informative but are conservative and require handling the nuisance concentration parameter. LIML reduces bias but does not eliminate all inference problems. The safest reading is therefore not to search for a single mechanical fix, but to combine first-stage diagnostics, robust intervals, and transparent sensitivity analysis.